ISS 2103 • EXAM REVISION PACK 04

Data Link Layer: MAC, Error & Flow Control, LAN Devices

Course outline weeks 3–4: medium access sub-layer, channel allocation, ALOHA, IEEE 802, data link protocols, error handling, bridges & switches

Priority: **VERY HIGH**

This topic is in **every** paper: 26 question parts worth 3–10 marks each. CSMA/CD is in 3 papers, LLC/MAC in 2, and the 2025 paper spent 23 marks on bridges, switches and routers. Most marks are for clear definitions, a comparison table and a labelled diagram, so learn the pictures in this pack.

How often this topic is examined

Question asked	Paper(s)	Marks
CSMA/CD: define / explain with a diagram / in Ethernet	2018 Q3b(i), SE2024 Q2b, ISA2024 Q4a	3–5
CSMA/CA; table CSMA/CD vs CSMA/CA	2018 Q3b(ii), 2023 Q4a	3–5
LLC and MAC sub-layers	2023 Q5a, ISA2024 Q1c	6
Pure and slotted ALOHA with diagrams	ISA2024 Q4c	4+4
Error types + detection methods	2023 Q4b, SE2024 Q3a	3–4
FEC, Hamming distance, piggybacking	2018 Q1b	3 each
Stop-and-wait protocol	2018 Q4b	4
Structure of a MAC address	2018 Q4d	5
Corrupted destination address on a bus LAN	2025 Q5b	3+3
Bridge in data-link switching; repeater vs bridge	2025 Q4c, SE2024 Q1b(ii)	2–7
Hub vs switch; why a switch is intelligent	SE2024 Q4f, 2018 Q5d	3–5
Router vs switch (layers and functions)	2025 Q5d	10
VLAN and its uses	2018 Q5c	4
Also here (tagged T05 in the bank): ARP; network segmentation	2018 Q3b(iii), ISA2024 Q1b	3–6

EXAM TIP

Three diagrams pay again and again: the **CSMA/CD flowchart**, the **ALOHA timelines** and the **hub vs switch vs router** domain picture. For any “differentiate” question, draw a table with a *Basis* column: it is quicker to write and easier to mark.

1 The data link layer and its two sub-layers

The data link layer (Layer 2) gives **node-to-node (hop-to-hop) delivery** of **frames** over one link. Its functions: **framing**, **physical (MAC) addressing**, **flow control**, **error control** and **access control** (who may use a shared medium). IEEE 802 splits it into two sub-layers.

- **Services to the network layer**: unacknowledged connectionless (Ethernet), acknowledged connectionless (Wi-Fi), acknowledged connection-oriented (numbered frames, used on unreliable links).
- **Node-to-node vs end-to-end**: Layer 2 delivers a frame to the next hop only; Layer 4 (TCP) gives end-to-end reliability.

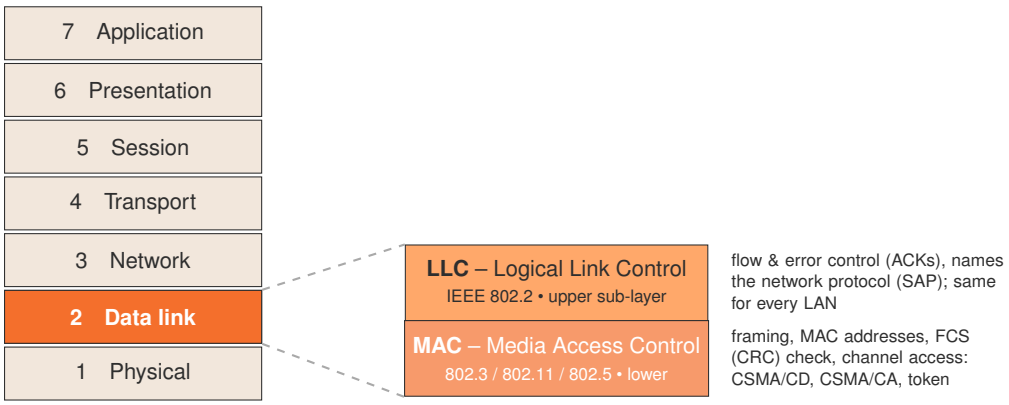


Figure: LLC sits on top of MAC inside the OSI data link layer. LLC talks to Layer 3, MAC talks to Layer 1.

PAST PAPER **2023 Q5a / ISA2024 Q1c** **[3+3 / 6 marks]**

The data link layer in the IEEE standard is divided into two (2) sub layers. Discuss the functions performed by: (i) LLC (ii) MAC. / Explain the layers in detail.

Model answer

(i) LLC – Logical Link Control (upper sub-layer, IEEE 802.2)

1. **Interface to the network layer**: identifies which Layer 3 protocol (IPv4, IPv6, . . .) is carried, using service access points (SAPs), so several protocols can share one link (multiplexing).
2. **Flow control**: stops a fast sender swamping a slow receiver (stop-and-wait, sliding window).
3. **Error control**: sequence numbers, acknowledgements and retransmission of lost or damaged frames (in connection-oriented LLC).
4. Gives connectionless or connection-oriented service and is **independent of the medium**: the same LLC runs over Ethernet, Wi-Fi or Token Ring.

(ii) MAC – Media Access Control (lower sub-layer, 802.3 / 802.11)

1. **Framing**: builds the frame (preamble, destination and source MAC, type, data, FCS) and finds frame boundaries.
2. **Physical addressing**: adds 48-bit source and destination MAC addresses; accepts only frames addressed to the station (or broadcast).
3. **Media access control**: decides who may transmit on a shared medium and when: CSMA/CD (Ethernet), CSMA/CA (Wi-Fi), token passing.
4. **Error detection**: computes and checks the CRC in the FCS; damaged frames are discarded.
5. Interfaces with the physical layer, so it is **specific to each LAN technology**.

Easy split to remember: LLC = **logical** jobs (talk to Layer 3, flow, errors). MAC = **media** jobs (frame, address, access, CRC).

2 Framing, MAC addresses and ARP

2.1 Framing

Framing breaks the bit stream into frames so the receiver knows where each one starts and ends.

- **Character count**: a header field gives the frame length. One corrupted count loses synchronisation.
- **Flag bytes with byte stuffing**: frames start and end with a FLAG byte; a FLAG or ESC inside the data is preceded by an ESC byte.
- **Flag bits with bit stuffing** (HDLC): flag = `01111110`. After **five 1s** in the data the sender inserts a **0**; the receiver removes it. Data `01111110` is sent as `011111010`.
- **Physical-layer coding violations**: use a signal pattern that never appears in data (e.g. in Manchester coding).

2.2 The Ethernet (IEEE 802.3) frame

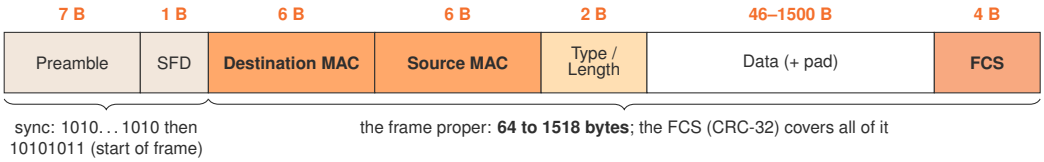


Figure: Ethernet II frame. Type 0x0800 = IPv4, 0x0806 = ARP, 0x86DD = IPv6. Short data is padded up to 46 bytes.

2.3 The MAC address

PAST PAPER **2018 Q4d** **[5 marks]**

With the aid of an example describe the structure of MAC address.

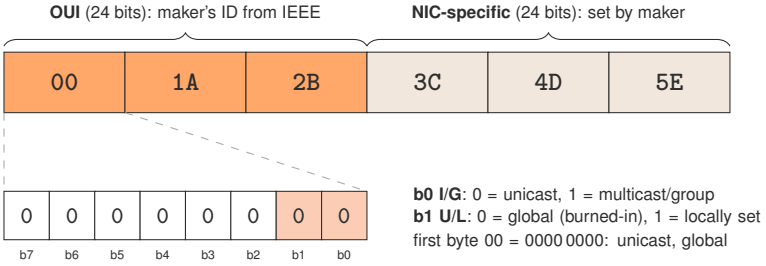


Figure: a 48-bit MAC address 00-1A-2B-3C-4D-5E and the two special bits of its first byte.

Model answer

1. A MAC address is a **48-bit (6-byte) physical address**, written as 12 hex digits, e.g. `00-1A-2B-3C-4D-5E`. It is burned into the NIC's ROM and used for delivery **inside one LAN**.
2. **First 24 bits = OUI** (Organisationally Unique Identifier): the manufacturer's code, assigned by IEEE (`00-1A-2B` above).
3. **Last 24 bits = NIC-specific**: a serial number the manufacturer gives each card (`3C-4D-5E`), so every address is globally unique.
4. In the first byte, bit b0 (I/G) says **unicast or multicast** and bit b1 (U/L) says **globally or locally administered**. The **broadcast** address is all 1s: `FF-FF-FF-FF-FF-FF`.
5. Written `00-1A-2B-3C-4D-5E` (Windows, see `ipconfig /all`), `00:1A:2B:3C:4D:5E` (Linux) or `001A.2B3C.4D5E` (Cisco).

2.4 ARP: from IP address to MAC address

MAC = **physical**, flat, 48 bits, changes nothing when you move. IP = **logical**, hierarchical, 32 bits, changes with the network. 2025 Q5c asks exactly this.

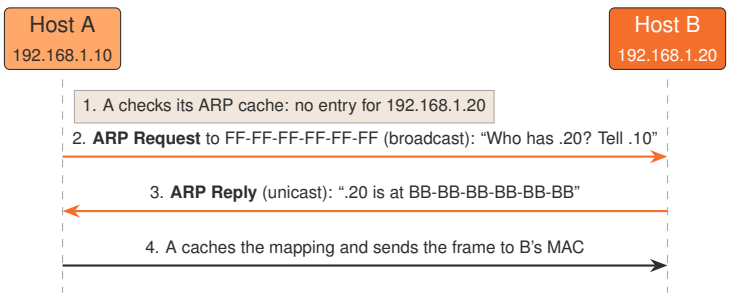


Figure: ARP. Every host in the LAN hears the request; only the owner of the IP replies.

PAST PAPER **2018 Q3b(iii)** **[3 marks]**

Define the following term: ARP.

Model answer

ARP (Address Resolution Protocol) finds the **MAC address** that belongs to a known **IP address** on the local network. The sender broadcasts an ARP Request ("who has this IP?"); the owner sends back a unicast ARP Reply with its MAC; the sender stores it in its **ARP cache** (`arp -a`) and can now build the frame. For a host on another network, ARP asks for the MAC of the **default gateway**. (RARP does the reverse: MAC → IP.)

2.5 When the destination address is corrupted

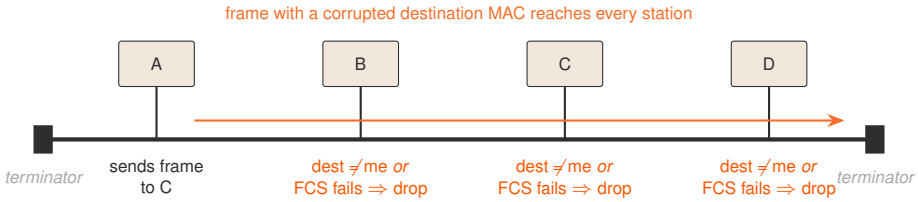


Figure: on a bus every station sees the frame, but nobody accepts it.

PAST PAPER **2025 Q5b(i) / Q5b(ii)** **[3+3 marks]**

Suppose a computer sends a frame to another computer on a bus topology LAN. The physical destination address of the frame is corrupted during the transmission: (i) Explain what happens to the frame? (ii) Explain how can the sender be informed about the situation?

Model answer

(i) On a bus the frame travels to **every station**. Each NIC compares the destination MAC with its own. The intended receiver sees an address that is not its own, so it **ignores the frame**. Even if the corrupted address matches another station, that station recalculates the CRC; the FCS covers the address field, so the **CRC check fails and the frame is discarded**. The frame is lost and is absorbed by the terminators at the ends of the bus.
(ii) Ethernet sends **no error message** for a discarded frame, and the sender saw no collision, so Layer 2 thinks it succeeded. The sender finds out **indirectly**: no **acknowledgement** comes back. In an ARQ protocol (stop-and-wait, sliding window in LLC) or in TCP at Layer 4, the sender's **timer expires**, it assumes the frame was lost and **retransmits** it. (A NAK can only be sent if the receiver recognises the frame as damaged and its own.)

3 Channel allocation and ALOHA

When many stations share one channel, the MAC sub-layer must allocate it.

- **Static allocation** (FDM, TDM): each user gets a fixed share. Wasteful for bursty LAN traffic, because idle shares are lost.
- **Dynamic allocation**: **random access** (ALOHA, CSMA, CSMA/CD, CSMA/CA), **controlled access** (reservation, polling, token passing) or **channelisation** (FDMA, TDMA, CDMA).

3.1 Pure and slotted ALOHA

ALOHA was built at the University of Hawaii (1970) to link islands by radio. It is the simplest random-access protocol.

- **Pure ALOHA**: send whenever you have a frame, with no carrier sense. Wait for an ACK; no ACK in time = collision, so wait a **random back-off** and send again.
- **Slotted ALOHA**: time is divided into **slots of one frame time** T_{fr}; stations may only start at the **beginning of a slot** (needs clock synchronisation). A frame collides only with frames in the same slot.

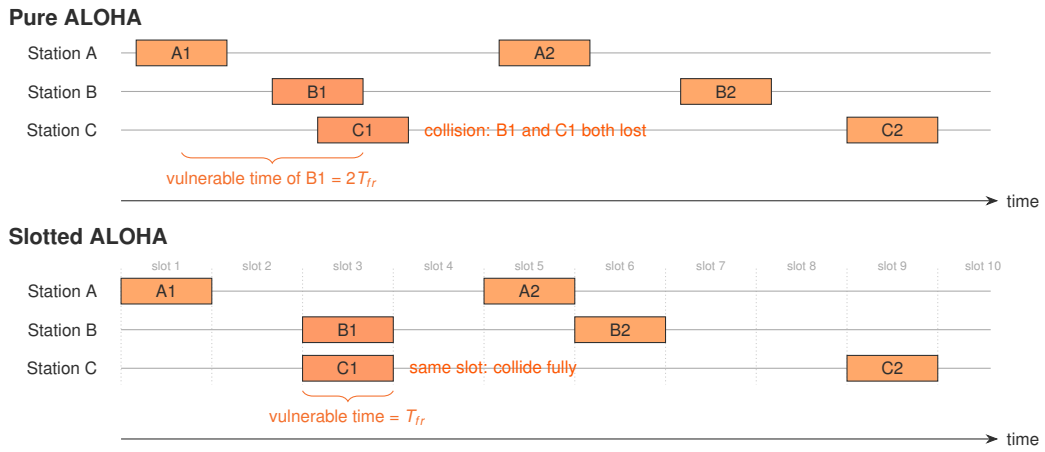


Figure: pure ALOHA frames start at any time and can overlap partly; slotted ALOHA frames start only on slot boundaries. Red = collided.

3.2 Throughput maths

G = average number of frames *offered* (new + retransmitted) per frame time. S = average number of *successful* frames per frame time (the throughput).

ALOHA throughput

$$P(\text{no other frame in } k \text{ frame times}) = e^{-kG} \quad \leftarrow \textit{Poisson arrivals}$$

$$\text{Pure: } S = Ge^{-2G} \quad \leftarrow \textit{vulnerable } 2T_{fr}\textit{: } k = 2$$

$$\frac{dS}{dG} = e^{-2G}(1 - 2G) = 0 \Rightarrow G = \tfrac{1}{2} \quad \leftarrow \textit{maximise}$$

$$S_{max} = \tfrac{1}{2}e^{-1} = \tfrac{1}{2e} \approx 0.184 = 18.4\%$$

$$\text{Slotted: } S = Ge^{-G} \quad \leftarrow \textit{vulnerable } T_{fr}\textit{: } k = 1$$

$$\frac{dS}{dG} = e^{-G}(1 - G) = 0 \Rightarrow G = 1$$

$$S_{max} = e^{-1} \approx 0.368 = 36.8\% \quad \leftarrow \textit{double pure ALOHA}$$

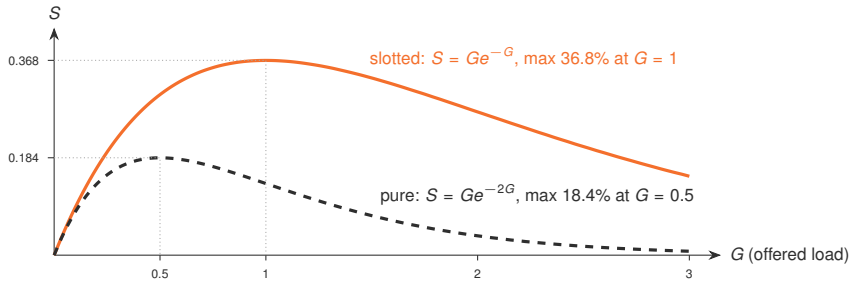


Figure: throughput rises with load, peaks, then falls as collisions take over.

Worked example: 200-bit frames on a 200 kbps shared channel

$$T_{fr} = 200/200\,000 = 1 \text{ ms} \quad \leftarrow \textit{so 1000 frame times per second}$$

$$1000 \text{ frames/s} \Rightarrow G = 1 \quad \leftarrow \textit{offered load}$$

$$\text{pure } S = 1 \cdot e^{-2} = 0.135 \Rightarrow 135 \text{ frames/s survive}$$

$$\text{slotted } S = 1 \cdot e^{-1} = 0.368 \Rightarrow 368 \text{ frames/s} \quad \leftarrow \textit{slotted at its peak}$$

$$500 \text{ frames/s} \Rightarrow G = 0.5 : \text{ pure } S = 0.5e^{-1} = 0.184 \Rightarrow 92 \quad \leftarrow \textit{pure at its peak}$$

PAST PAPER **ISA2024 Q4c(i) / Q4c(ii)** **[4+4 marks]**

In the Aloha system, the basic idea is applicable to any system in which uncoordinated users are competing for the use of a single shared channel. Explain using a diagram the functionality of: (i) Pure Aloha (ii) Slotted Aloha.

Model answer

Draw the two timelines above (A1, B1/C1 collision, slots).

(i) Pure ALOHA

1. A station **transmits whenever it has a frame**; there is no carrier sensing and no time slots.
2. It waits for an **ACK** for a time-out of $2\times$ the maximum propagation delay. No ACK $\Rightarrow$ the frame collided.
3. After a collision it waits a **random back-off** time and retransmits (random, so the colliding stations do not collide again).
4. Any overlap destroys both frames, so the **vulnerable time is** $2T_{fr}$ and the **maximum throughput is 18.4%** ($S = Ge^{-2G}$ at $G = 0.5$).

(ii) Slotted ALOHA

1. Time is divided into **slots equal to one frame time**; all stations are synchronised to a clock.
2. A station may send **only at the start of a slot**; a frame created mid-slot waits for the next slot.
3. Collisions happen only when two stations pick the **same slot**; they are then resent after a random number of slots.
4. The **vulnerable time halves to** T_{fr}, so the **maximum throughput doubles to 36.8%** ($S = Ge^{-G}$ at $G = 1$). The price is the need for synchronisation.

Write both numbers with their formula: 18.4% = $1/(2e)$ and 36.8% = $1/e$. Examiners love the "vulnerable time" phrase.

4 CSMA, CSMA/CD and CSMA/CA

CSMA (Carrier Sense Multiple Access) improves on ALOHA: **listen before talk**. Collisions still happen because of **propagation delay**: a station may sense the line idle while another's signal is still on its way.

- **1-persistent**: sense continuously; send as soon as the line is idle (Ethernet). Many collisions if two are waiting.
- **Non-persistent**: if busy, wait a random time, then sense again. Fewer collisions, more delay.
- **p-persistent** (slotted channels): when idle, send with probability p, otherwise wait one slot.

4.1 CSMA/CD: collision detection (wired Ethernet, IEEE 802.3)

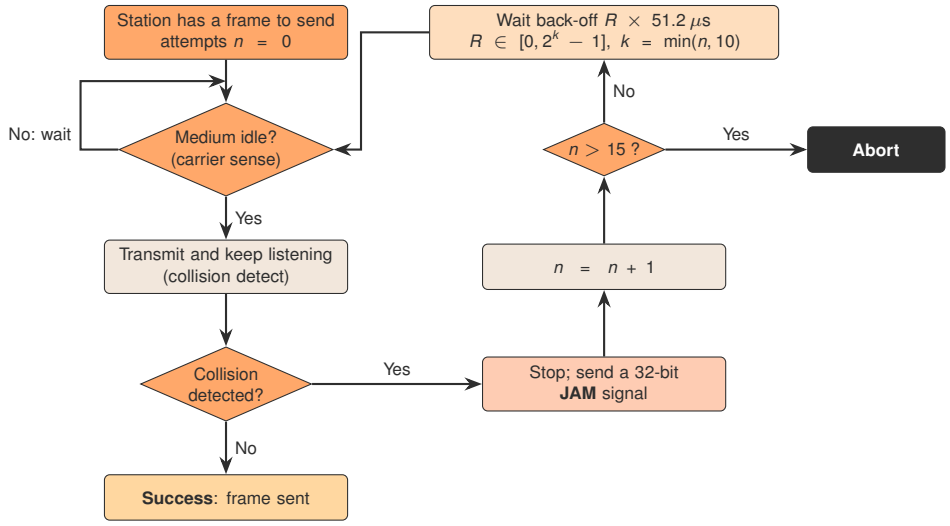


Figure: CSMA/CD flowchart (binary exponential back-off, 16 attempts maximum).

Why Ethernet frames are at least 64 bytes

$$T_{fr} \geq 2\,T_p \qquad \leftarrow \textit{still sending when collision returns}$$
$$L_{min} = \text{bandwidth} \times 2T_p = 10 \times 10^6 \times 51.2 \times 10^{-6} \qquad \leftarrow \textit{10 Mbps, round trip } 51.2\,\mu s$$
$$= 512 \text{ bits} = 64 \text{ bytes}$$

PAST PAPER **ISA2024 Q4a / SE2024 Q2b / 2018 Q3b(i)** **[3–5 marks]**

With the aid of a supporting diagram, explain the concept of Carrier Sense Multiple Access/Collision Detection mechanism used by Ethernet (CSMA/CD). / In the description of the Ethernet Architecture, explain the function and operation of the CSMA/CD flow control paradigm. / Define CSMA/CD.

Model answer

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is the MAC method of classic half-duplex Ethernet (IEEE 802.3) that lets many stations share one cable (bus or hub). Draw the flowchart above.

1. **Carrier sense**: a station listens before sending; if the medium is busy it waits (1-persistent).
2. **Multiple access**: when the medium is idle any station may transmit.
3. **Collision detection**: while sending it keeps listening; if the signal on the wire differs from what it sent (voltage too high), a collision has occurred.
4. On a collision it **stops** and sends a **jam signal** so every station knows.
5. It waits a random **binary exponential back-off**: $R\times$ slot time, R from 0 to $2^k - 1$; after 16 failed attempts it gives up.
6. A frame must be at least **64 bytes** so the sender is still transmitting when a collision anywhere on the LAN reaches it.

In modern **full-duplex switched** Ethernet there are no collisions, so CSMA/CD is switched off.

4.2 CSMA/CA: collision avoidance (Wi-Fi, IEEE 802.11)

For the 3-mark "define" (2018) the first three points are enough: listen before, many share, listen while sending and back off.

A wireless station **cannot detect collisions**: its own strong signal drowns weak incoming ones, and of the **hidden station problem**. So it tries to **avoid** collisions instead.



Figure: the hidden station problem. RTS/CTS solves it because C hears B's CTS.

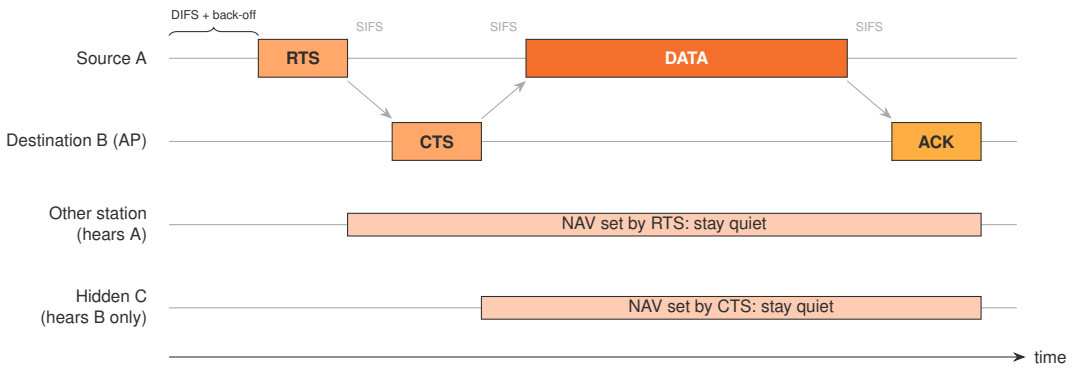


Figure: CSMA/CA with RTS/CTS. Everyone who hears the RTS or the CTS defers (NAV) until the ACK is over.

1. Sense the channel; when idle, wait a gap called **DIFS**, then a **random back-off** (the counter only runs while the channel is idle).

2. Optionally send a short **RTS** (Request To Send) with the time needed; the AP answers **CTS** (Clear To Send). All others set their **NAV** timer and stay quiet.
3. Send the DATA; the receiver waits **SIFS** and returns an **ACK**. No ACK ⇒ assume a collision, double the contention window and retry.

PAST PAPER **2018 Q3b(ii)** **[3 marks]**

Define the following term: CSMA/CA.

Model answer

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) is the MAC method of wireless LANs (IEEE 802.11). A station senses the channel and, when it is idle, waits an inter-frame space (DIFS) plus a random back-off before sending, so that collisions are **avoided before they happen**. It may reserve the channel with **RTS/CTS** (others defer using the NAV), and every frame must be confirmed by an **ACK**; no ACK means retransmit.

PAST PAPER **2023 Q4a** **[5 marks]**

Using a table, differentiate CSMA/CD from CSMA/CA.

Model answer

Basis	CSMA/CD	CSMA/CA
Meaning	Collision **Detection**: deals with a collision *after* it happens	Collision **Avoidance**: tries to prevent it *before* it happens
Network	Wired Ethernet, IEEE 802.3	Wireless LAN (Wi-Fi), IEEE 802.11
Listening	Listens *while* transmitting	Cannot listen while sending; uses waiting times (IFS) and back-off
On collision	Stop, send jam signal, binary exponential back-off	No detection; a missing ACK means retry with a larger window
ACK	No ACK at the MAC layer	Positive ACK for every frame
Extra mechanism	Minimum frame size (64 bytes)	RTS/CTS handshake and NAV (fixes hidden stations)
Efficiency	Low overhead, efficient on cable	More overhead (IFS, RTS/CTS, ACK)

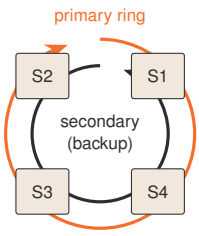
REMEMBER IT

CD = cable, detect, after. CA = air, avoid, before.

5 Controlled access, IEEE 802 standards and FDDI

Controlled access removes collisions: **reservation** (book a slot first), **polling** (a primary asks each secondary in turn) and **token passing** (only the holder of a special frame, the token, may send: Token Bus 802.4, Token Ring 802.5, FDDI).

Standard	Covers
802.1	Bridging, STP (802.1D), VLANs (802.1Q)
802.2	LLC sub-layer
802.3	Ethernet (CSMA/CD)
802.4	Token Bus
802.5	Token Ring
802.11	Wireless LAN, Wi-Fi (CSMA/CA)
802.15	WPAN: Bluetooth, Zigbee
802.16	WiMAX (wireless MAN)



FDDI: two counter-rotating fibre rings.

FDDI – Fibre Distributed Data Interface (ANSI standard, taught with IEEE 802)

A **100 Mbps** token-passing LAN/MAN over **optical fibre**, built as **two counter-rotating rings**. Normally only the primary ring carries data; if a cable or station fails, the stations next to the break **wrap** onto the secondary ring, so the network keeps working (**fault tolerance**). Ring up to 100 km, up to 500 stations; used as a campus backbone before Gigabit Ethernet.

6 Error detection and correction

- **Single-bit error**: only one bit of the data unit changes, e.g. 0100110111 → 0000110111. Rare in serial links.
- **Burst error**: **two or more** bits change (not necessarily next to each other). The **burst length** is from the first to the last corrupted bit. Common, because noise lasts longer than one bit time.
- All methods add **redundant bits**. **Detection** only says "something is wrong" (then ARQ: retransmit). **Correction** finds and fixes the bit (FEC).

6.1 Parity: simple (VRC) and two-dimensional (LRC)

Add one bit so the number of 1s is **even** (even parity) or odd. 1100001 has three 1s, so even parity sends 11000011. Detects every single-bit error and any *odd* number of errors; it misses two flipped bits. **2-D parity** arranges the data in rows, adds a parity bit to each row *and* a parity row for each column:

data (7 bits)	row parity
1 1 0 0 1 1 1	1
1 0 1 1 1 0 1	1
0 1 1 1 0 0 1	0
0 1 0 1 0 0 1	1
0 1 0 1 0 1 0	1

Bottom row = column parity. A single flipped bit shows up as one bad row *and* one bad column, which pin-points it.

6.2 CRC: cyclic redundancy check (the strongest)

1. Sender and receiver agree a **generator (divisor)** of $n+1$ bits, e.g. 1101 = $x^3 + x^2 + 1$ ($n = 3$).
2. Append n zeros to the data and divide by the generator using **modulo-2 division** (XOR, no carries).
3. The n-bit **remainder is the CRC**; it replaces the zeros. The codeword is now exactly divisible by the generator.
4. The receiver divides the whole codeword by the same generator: remainder **0 = accept**, anything else = **reject**.

```
                1 1 1 1 0 1     quotient (thrown away)
1101 ) 1 0 0 1 0 0 0 0 0        data 100100 + three 0s
       1 1 0 1                  1: leading 1 ⇒ XOR 1101
       1 0 0 0 0                2: remainder 100, bring down 0
       1 1 0 1
         1 0 1 0 0              3: 1000 ⊕ 1101 = 101, bring down 0
         1 1 0 1
           1 1 1 0 0            4: 1010 ⊕ 1101 = 111, bring down 0
           1 1 0 1
             0 1 1 0 0          5: 1110 ⊕ 1101 = 011, bring down 0
             0 0 0 0 0            leading 0 ⇒ XOR 0000
               1 1 0 0 0        6: 0110 ⊕ 0000 = 110, bring down 0
               1 1 0 1            1100 ⊕ 1101
                 0 0 1 0 0      remainder 001 = CRC
```

Figure: modulo-2 long division of 100100 000 by 1101. Transmitted codeword = 100100 **001**.

Check at the receiver

$$\text{received } 100100\,001 \div 1101 \Rightarrow \text{remainder } 000 \quad \leftarrow \textit{accept}$$
$$\text{received } 100000\,001 \div 1101 \Rightarrow \text{remainder} \neq 000 \quad \leftarrow \textit{one bit flipped: reject}$$

A CRC with an n-bit remainder detects **all single-bit errors**, all odd numbers of errors and **all bursts of length** $\leq n$. Ethernet's FCS is CRC-32.

Modulo-2 rules: $1 \oplus 1 = 0$, $1 \oplus 0 = 1$, no borrows. If the leading bit of the window is 0, XOR with 0000. The CRC always has one bit fewer than the divisor.

6.3 Checksum

The data is cut into k-bit words which are added in **1's complement** arithmetic (a carry out of the top is wrapped round and added back). The **complement of the sum** is sent as the checksum. The receiver adds all words plus the checksum: all 1s (complement = 0) means no error. Used in the IP, TCP and UDP headers.

Worked example: two 8-bit words

$$10101001 + 00111001 \quad \leftarrow \textit{169 + 57}$$
$$\text{sum} = 11100010 \quad \leftarrow \textit{226, no carry out}$$
$$\text{checksum} = \overline{11100010} = 00011101 \quad \leftarrow \textit{flip every bit (29)}$$
$$\text{receiver: } 10101001 + 00111001 + 00011101 = 11111111 \quad \leftarrow \textit{complement = 0, accept}$$

PAST PAPER **2023 Q4b** **[4 marks]**

Copy and complete Table 2 (Errors): Type of error | Description | Detection method, for: Single bit, Burst error.

Model answer

Type of error	Description	Detection method
Single bit	Only **one bit** of the data unit changes, 0 to 1 or 1 to 0, e.g. 00000010 → 00001010.	**Parity check** (VRC): one parity bit catches every single-bit error (CRC and checksum also work).
Burst error	**Two or more bits** change, not necessarily adjacent; length = first to last bad bit. Caused by noise longer than a bit.	**CRC**: catches all bursts up to the CRC length (also 2-D parity, checksum).

PAST PAPER **SE2024 Q3a** **[3 marks]**

Explain any three error detection strategy used to control data transmission.

Model answer

1. **Parity check**: one extra bit makes the count of 1s even (or odd); the receiver recounts and a mismatch means an error. Detects single-bit and odd-numbered errors; 2-D parity also catches many bursts.
2. **CRC**: the sender divides the data (with n zeros added) by an agreed generator using modulo-2 division and appends the n-bit remainder. The receiver divides again: remainder 0 = no error. Very strong; used in Ethernet.
3. **Checksum**: the data words are added in 1's complement and the complement of the sum is sent. The receiver's sum of everything must be all 1s. Used in IP, TCP, UDP.

6.4 Hamming distance, Hamming code and FEC

Hamming distance $d(x, y)$ = number of positions where two equal-length words differ = number of 1s in $x \oplus y$.

Hamming distance and what d_{min} buys you

$$d(10101, 11110) : 10101 \oplus 11110 = 01011 \Rightarrow d = 3$$

detect up to s errors $\iff$ $d_{min} \geq s + 1$

correct up to t errors $\iff$ $d_{min} \geq 2t + 1$ ← *so $d_{min} = 3$ corrects 1 error*

Hamming code places r parity bits at positions 1, 2, 4, 8, ... with $2^r \geq m + r + 1$ (m = data bits). Each parity bit covers the positions whose binary number contains it: r1 → 1,3,5,7,9,11; r2 → 2,3,6,7,10,11; r4 → 4–7; r8 → 8–11.

11	10	9	8	7	6	5	4	3	2	1
1	0	0	1	1	1	0	0	1	0	1
d	d	d	r8	d	d	d	r4	d	r2	r1

data 1001101 (7 bits) needs $r = 4$: $2^4 = 16 \geq 12$

Figure: Hamming(11,7) codeword 10011100101 for data 1001101 (even parity). Peach = parity bits.

Correcting a single-bit error

r1 = parity of bits 3, 5, 7, 9, 11 = 1, 0, 1, 0, 1 ⇒ 1 ← *three 1s, make it even*

r2 = bits 3, 6, 7, 10, 11 = 1, 1, 1, 0, 1 ⇒ 0

r4 = bits 5, 6, 7 = 0, 1, 1 ⇒ 0

r8 = bits 9, 10, 11 = 0, 0, 1 ⇒ 1

received 10010100101 ← *bit 7 flipped*

recheck r8 r4 r2 r1 = 0111 ← *the syndrome*

$0111_2 = 7$ ⇒ flip bit 7 back ← *error found and corrected*

PAST PAPER **2018 Q1b(a)** **[3 marks]**

Explain the meaning of the following term: Forward error correction (FEC).

Model answer

FEC means the sender adds enough **redundant bits** (e.g. a Hamming or Reed-Solomon code) that the **receiver can find and correct errors by itself**, without asking for a retransmission. It is used when retransmission is impossible or too slow: real-time voice/video, satellite and deep-space links, simplex (one-way) broadcast. The opposite is **ARQ** (Automatic Repeat reQuest): only detect the error and ask for the frame again.

PAST PAPER **2018 Q1b(b)** **[3 marks]**

Explain the meaning of: Hamming distance.

Model answer

The **Hamming distance** between two codewords of equal length is the **number of bit positions in which they differ**. Find it by XORing them and counting the 1s: $d(10101, 11110)$: $10101 \oplus 11110 = 01011$, so $d = 3$. The **minimum** Hamming distance of a code sets its power: it detects s errors if $d_{min} \geq s + 1$ and corrects t errors if $d_{min} \geq 2t + 1$.

7 Flow and error control: stop-and-wait and sliding windows

Flow control stops a fast sender overrunning a slow receiver. **Error control** (ARQ) uses sequence numbers, ACKs, timers and retransmission to recover lost or damaged frames.

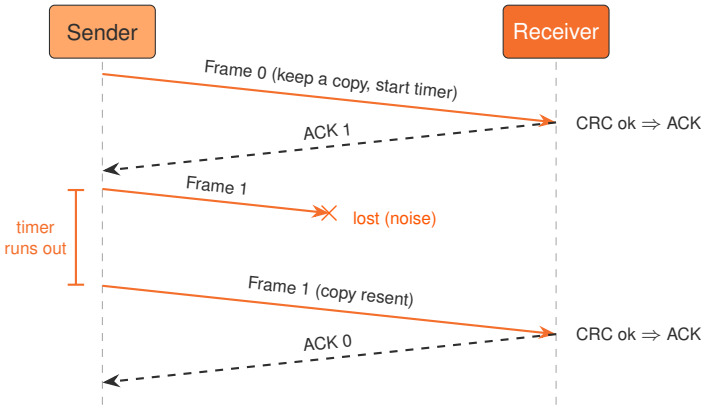


Figure: stop-and-wait ARQ with a lost frame. Frames and ACKs are numbered 0, 1, 0, 1 ... (ACK = number of the next frame expected).

PAST PAPER **2018 Q4b** **[4 marks]**

Explain the working principle of stop and wait protocol.

Model answer

1. The sender transmits **one frame**, keeps a copy and starts a **timer**, then **stops and waits**.
2. The receiver checks the frame (CRC); if it is good it passes it up and returns an **ACK**. Only when the ACK arrives does the sender send the next frame (window size = 1).
3. If the frame or the ACK is lost or damaged, the **timer expires** and the sender **retransmits** its copy (stop-and-wait ARQ).
4. Frames are numbered **0 and 1 alternately** so the receiver can spot and discard a duplicate. Simple, but the line is idle while waiting, so it is inefficient on long or fast links.

How inefficient? Link utilisation of stop-and-wait

$$a = \frac{T_p}{T_{fr}} \quad \leftarrow \textit{propagation time in frame times}$$

$$U = \frac{T_{fr}}{T_{fr} + 2T_p} = \frac{1}{1 + 2a}$$

1 Mbps, 1000-bit frames: $T_{fr} = 1$ ms; $T_p = 10$ ms $\Rightarrow a = 10$

$$U = \tfrac{1}{1+20} \approx 4.8\% \quad \leftarrow \textit{idle 95\% of the time}$$

to fill the pipe, window $W \geq 1 + 2a = 21$ frames

7.1 Sliding window: Go-Back-N and Selective Repeat

The sender may send up to W frames before it needs an ACK. With m-bit sequence numbers (0 to $2^m - 1$) the window slides forward as ACKs arrive.

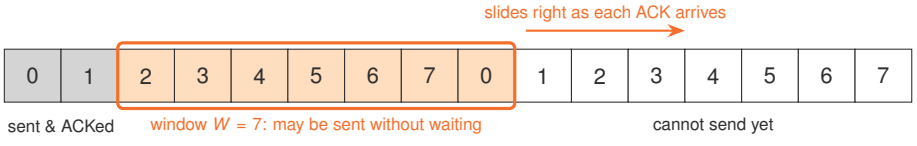


Figure: sender's sliding window with 3-bit sequence numbers (Go-Back-N maximum $W = 2^3 - 1 = 7$).

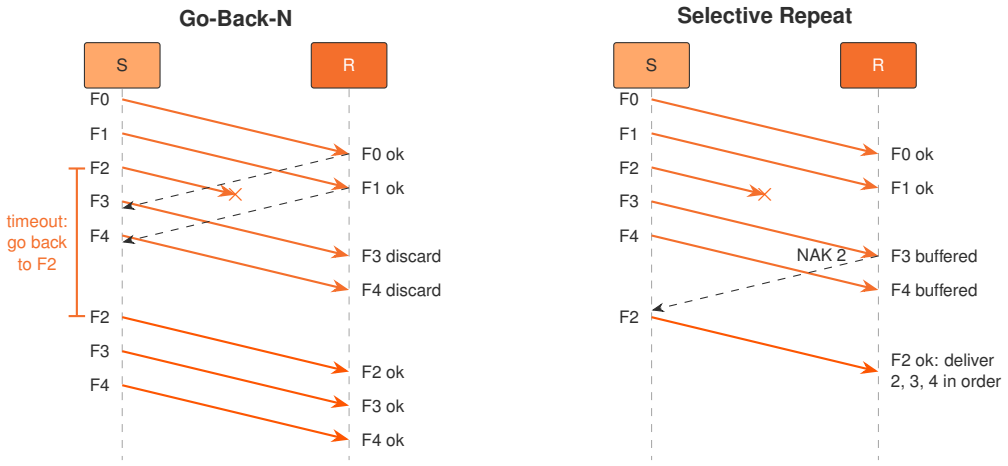


Figure: frame 2 is lost. Go-Back-N resends 2, 3 and 4; Selective Repeat resends only 2. Red = retransmissions.

	Stop-and-wait	Go-Back-N	Selective Repeat
Sender window	1	up to $2^m - 1$	up to 2^{m-1}
Receiver window	1	1 (accepts only in order)	up to 2^{m-1} (buffers)
On a lost frame	resend it	resend it **and all after it**	resend **only that frame**
Good for	short links	cheap receivers, low error rate	noisy, long links; complex

PAST PAPER **2018 Q1b(c)** **[3 marks]**

Explain the meaning of: Piggybacking.

Window limits are a favourite trap: GBN $2^m - 1$, SR 2^{m-1}. With $m = 3$: GBN 7, SR 4.

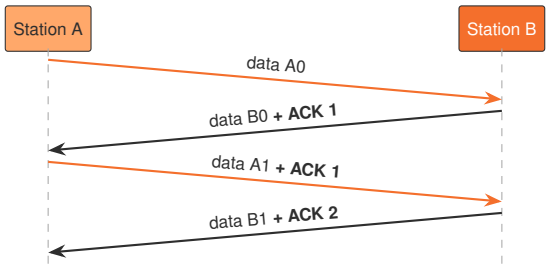


Figure: the ACK rides in the header of the next data frame going the other way.

Model answer

In **two-way (full-duplex)** communication the receiver does not send a separate ACK frame; it **attaches the acknowledgement to its next outgoing data frame** (an ACK field in the header). **Advantage**: fewer frames, so better use of bandwidth and less overhead. **Disadvantage**: if the station has no data to send the ACK is delayed, so an ACK timer forces a separate ACK when it runs out.

8 LAN devices: repeater, hub, bridge, switch, router

Device	Layer	What it does	Collision domains	Broadcast domains
Repeater	1	Regenerates the signal bit by bit to extend distance	1 (shared)	1
Hub	1	Multiport repeater: copies every bit to every port	1 (shared)	1
Bridge	2	Filters/forwards frames between segments by MAC address	1 per port	1
Switch	2	Multiport bridge in hardware; full duplex per port	1 per port	1 (per VLAN)
Router	3	Forwards packets between networks by IP address	1 per port	1 per interface

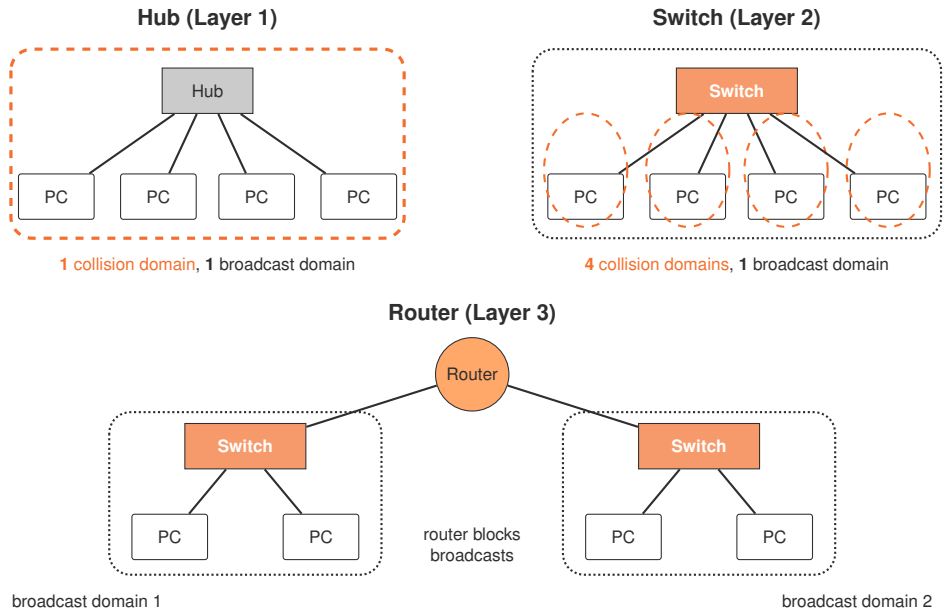
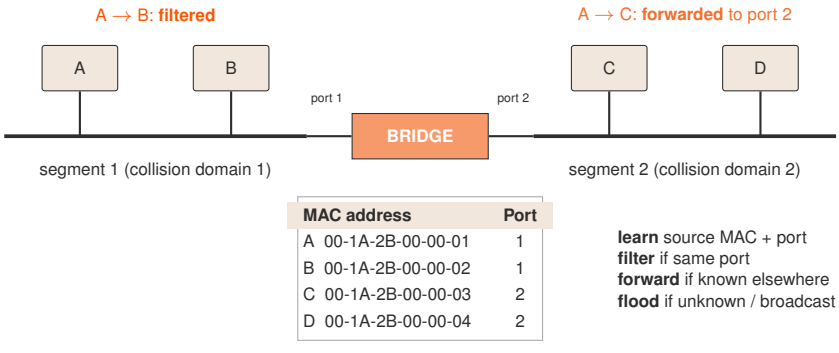


Figure: dashed orange = collision domain, dotted grey = broadcast domain. Switches split collision domains; routers split broadcast domains.

8.1 How a bridge (and a switch) learns



MAC address	Port
A 00-1A-2B-00-00-01	1
B 00-1A-2B-00-00-02	1
C 00-1A-2B-00-00-03	2
D 00-1A-2B-00-00-04	2

learn source MAC + port
filter if same port
forward if known elsewhere
flood if unknown / broadcast

Figure: a transparent (learning) bridge and its MAC table.

Bridges learn from the **source** address and forward on the **destination** address. Mixing these up loses the mark.

PAST PAPER **2025 Q4c** **[7 marks]**

Describe the function of a network bridge and how it operates in a Data Link Layer switching environment.

Model answer

A **bridge** is a **Layer 2** device that joins two or more LAN segments and **filters and forwards**

frames using MAC addresses. Each segment becomes its **own collision domain**, but the whole network stays **one broadcast domain**. Draw the figure above.

1. **Learning**: for each arriving frame it records the **source MAC** and the port it came in on, building its MAC (forwarding) table automatically (transparent bridge, no set-up).
2. **Filtering**: if the destination is on the **same port** the frame came from, it is dropped, so local traffic stays local.
3. **Forwarding**: if the destination is known on another port, the frame goes **only to that port**.
4. **Flooding**: unknown destinations, broadcasts and multicasts go out of every port except the incoming one.
5. **Ageing**: entries time out (about 300 s) so moved hosts are relearned.
6. **Loop prevention**: with redundant bridges, the **Spanning Tree Protocol** (STP, IEEE 802.1D) blocks some ports so frames cannot loop forever.
7. It checks the FCS and drops bad frames, and can join segments of different speeds. A **switch** is a multiport bridge doing all this in hardware (ASICs) at wire speed.

PAST PAPER **SE2024 Q1b(ii)** **[2 marks]**

Explain the functional difference between a Repeater and Bridge.

Model answer

A **repeater** (Layer 1) only **regenerates and re-times the signal** bit by bit to extend cable distance; it does not read addresses and passes on everything, collisions included (one collision domain). A **bridge** (Layer 2) **reads MAC addresses** and filters or forwards frames between segments, cutting traffic and **separating collision domains**.

PAST PAPER **SE2024 Q4f** **[5 marks]**

Differentiate between Hub & switch on the basis of layer, ports, device type, speed.

Model answer

Basis	Hub	Switch
Layer	Layer 1, Physical	Layer 2, Data link (Layer 3 switches exist)
Ports	Typically 4–12; all ports share **one collision domain**	8–48; **each port is its own collision domain**
Device type	Passive, "dumb" multiport repeater: sends every frame out of every port	Active, intelligent: uses a MAC table to send a frame only to the destination port
Speed	Shared bandwidth (e.g. 10/100 Mbps split among all), half duplex	Dedicated bandwidth per port (100 Mbps–10 Gbps), full duplex
Addressing / security	No addresses; anyone can sniff all traffic; many collisions	MAC addresses; traffic goes only where needed; no collisions in full duplex

PAST PAPER **2018 Q5d** **[3 marks]**

Explain why a switch is said to be intelligent.

Model answer

1. It **learns** the MAC addresses on each port and builds a MAC address table by itself.
2. It **makes a forwarding decision** for every frame: sends it only to the destination port and filters local traffic, instead of copying it everywhere like a hub.
3. It checks frames for errors (store-and-forward) and supports VLANs, full duplex, QoS, port security and STP; managed switches can be configured and monitored.

PAST PAPER **2025 Q5d** **[10 marks]**

Discuss the purpose of a router and a switch in a network. Which OSI layers do they operate on, and what is the difference between them in terms of functionality?

Model answer

Switch: connects end devices **within one LAN** and moves frames between them. Works at **Layer 2 (data link)**; a multilayer switch can also route at Layer 3. **Router**: connects **different networks** (LAN to LAN, LAN to WAN/Internet) and chooses the best path for packets. Works at **Layer 3 (network)**.

Basis	Switch	Router
Address used	MAC addresses; MAC address table	IP addresses; routing table
Data unit	Frames	Packets
Decision	Forwards a frame to the correct port (learn, filter, forward, flood)	Finds the best path using static routes or routing protocols (RIP, OSPF, EIGRP)
Broadcasts	Forwards broadcasts: one broadcast domain (unless VLANs)	Blocks broadcasts: each interface is a separate broadcast domain
Collision domains	One per port	One per port
Extra features	VLANs, STP, port security, PoE	NAT, DHCP, ACLs/firewall, VPN, WAN interfaces
Speed	Very fast, hardware (ASIC) switching	Slower: must read and rewrite the Layer 3 header, decrement TTL
Example	Lab switch joining 24 PCs	Router joining the campus LAN to the ISP

For 10 marks: purpose of each (2+2), layers (2), then at least four rows of differences. Draw the hub/switch/router picture if you have time.

9 VLANs and network segmentation

A **VLAN** (Virtual LAN) is a **logical group of switch ports that forms its own broadcast domain**, whatever the physical location of the devices. It is set in software on the switch. Frames between switches carry an **IEEE 802.1Q tag** on a **trunk** link; traffic between VLANs must go through a **router** or Layer 3 switch.

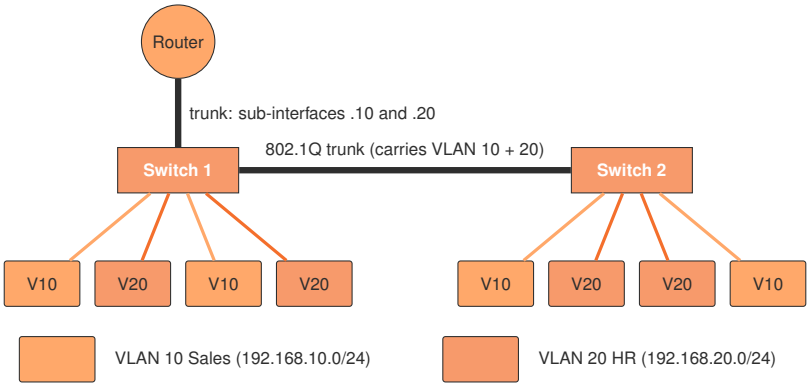


Figure: two VLANs spread over two switches. A broadcast from a VLAN 10 PC reaches only VLAN 10 PCs; VLAN 10 to VLAN 20 goes via the router ("router on a stick").



Figure: the 802.1Q tag is inserted after the source MAC. 12-bit VLAN ID gives up to 4094 VLANs.

PAST PAPER **2018 Q5c** **[4 marks]**

Define the term VLAN giving its uses in your answer.

Model answer

A **VLAN** is a logical grouping of devices on one or more switches into a **separate broadcast domain**, configured in software (IEEE 802.1Q), regardless of where the devices physically are.
Uses:

1. **Group departments** (Accounts, HR, IT) on shared switches as if they had their own LAN.
2. **Security**: sensitive traffic is isolated; other VLANs can reach it only through a router with ACLs.
3. **Performance**: smaller broadcast domains mean less broadcast traffic.
4. **Flexibility**: move a user by changing the port's VLAN, not the cabling; separate voice and data (voice VLAN with QoS).

PAST PAPER **ISA2024 Q1b** **[6 marks]**

Explain the concept of network segmentation.

Model answer

Network segmentation divides one large network into smaller **segments**, each with its own traffic and access rules, using switches/bridges (collision domains), **VLANs** and **subnets** with routers (broadcast domains), and firewalls (security zones such as a DMZ).

1. **Performance**: fewer devices share each collision and broadcast domain, so less congestion.
2. **Security**: an attack or malware is contained in one segment; it limits lateral movement; access control lists or firewalls filter traffic between zones.
3. **Management and troubleshooting**: faults stay inside one segment; the network is organised by department, floor or function.
4. **Compliance**: sensitive data (e.g. payment or HR systems) is kept in its own zone and can be audited.

Segmentation is in the T05 bank list but is answered with Layer 2 ideas (switches, VLANs) and Layer 3 ideas (subnets). Mention both.

10 Practice questions

PRACTICE Q1 **EASY** **[3 marks]**

A NIC has the MAC address 3C-A8-2A-11-05-FF. (a) Which part is the OUI? (b) Is it unicast or multicast? (c) Write the broadcast MAC address.

Answer

(a) OUI = `3C-A8-2A` (first 24 bits); NIC-specific part = `11-05-FF`. (b) First byte `3C` = `0011 1100`: bit b0 = 0, so **unicast** (and b1 = 0, globally administered). (c) `FF-FF-FF-FF-FF-FF`.

PRACTICE Q2 **EASY** **[2 marks]**

Find the even and the odd parity bit for the data `1011001`.

Answer

There are four 1s. Even parity: bit = **0** (sent `10110010`). Odd parity: bit = **1** (sent `10110011`).

PRACTICE Q3 **EASY** **[4 marks]**

Give the IEEE 802 standard number for: (a) Ethernet (b) Wi-Fi (c) VLAN tagging (d) the LLC sub-layer.

Answer
(a) 802.3 (b) 802.11 (c) 802.1Q (d) 802.2.

PRACTICE Q4 EASY [3 marks]

State the vulnerable time and the maximum throughput of pure ALOHA and of slotted ALOHA.

Answer
Pure: vulnerable time $2T_{fr}$, max throughput $1/(2e) = 18.4\%$ at $G = 0.5$. Slotted: vulnerable time T_{fr}, max throughput $1/e = 36.8\%$ at $G = 1$.

PRACTICE Q5 MEDIUM [4 marks]

Find the Hamming distance between `1011101` and `1001001`. What minimum Hamming distance must a code have to detect 3 errors? To correct 2 errors?

Answer

Working

$$1011101 \oplus 1001001 = 0010100 \Rightarrow d = 2 \qquad \leftarrow \textit{count the 1s}$$
$$\text{detect } s = 3 : d_{min} = s + 1 = 4$$
$$\text{correct } t = 2 : d_{min} = 2t + 1 = 5$$

PRACTICE Q6 MEDIUM [5 marks]

Compute the 8-bit checksum for the words `11001100` and `10101010`, and show how the receiver checks it.

Answer

Working

$$11001100 + 10101010 = 1\,01110110 \qquad \leftarrow \textit{204 + 170 = 374: carry out}$$
$$\text{wrap the carry: } 01110110 + 1 = 01110111 \qquad \leftarrow \textit{1's complement (119)}$$
$$\text{checksum} = \overline{01110111} = 10001000 \qquad \leftarrow \textit{136}$$
$$\text{receiver: } 204 + 170 + 136 = 510 = 1\,11111110 \qquad \leftarrow \textit{carry out again}$$
$$11111110 + 1 = 11111111 \qquad \leftarrow \textit{all 1s: accept}$$

PRACTICE Q7 MEDIUM [5 marks]

A group of stations shares a 200 kbps channel using 200-bit frames. Together they offer 500 frames per second. Find the throughput (frames/s) with pure ALOHA and with slotted ALOHA.

Answer

Working

$$T_{fr} = 200/200\,000 = 1 \text{ ms} \Rightarrow G = 500 \times 0.001 = 0.5 \qquad \leftarrow \textit{frames per frame time}$$
$$\text{pure } S = 0.5\,e^{-1} = 0.184 \Rightarrow 0.184 \times 1000 \approx 92 \text{ frames/s}$$
$$\text{slotted } S = 0.5\,e^{-0.5} = 0.303 \Rightarrow \approx 151 \text{ frames/s} \qquad \leftarrow \textit{1000 slots per second}$$

PRACTICE Q8 MEDIUM **[6 marks]**

Explain why the data link layer needs framing, describe two framing methods, and bit-stuff the data `0111111011111100`.

Answer

Framing lets the receiver find where each frame **starts and ends** in the bit stream, so it can check the CRC and pass whole frames up. (1) **Byte stuffing**: frames are delimited by FLAG bytes; any FLAG or ESC byte in the data gets an ESC byte in front. (2) **Bit stuffing**: the flag is `01111110`; after five consecutive 1s in the data the sender inserts a 0, which the receiver removes.

Working

$$0\ 11111\ 1\ 0\ 11111\ 1\ 00 \rightarrow 0\ 111110\ 1\ 0\ 111110\ 1\ 00 \quad \leftarrow \textit{a 0 after every five 1s}$$
$$= 0111110101111110100 \quad \leftarrow \textit{16 bits become 18}$$

PRACTICE Q9 HARD **[12 marks]**

Data `110101` is sent using CRC with the generator $x^3 + x + 1$. (a) Find the transmitted codeword. (b) Show that the receiver accepts it. (c) The 4th bit is flipped on the way. Show that the error is detected. (d) Which errors does this CRC always detect?

Answer

Working

$$x^3 + x + 1 = 1011 \Rightarrow n = 3,\ \text{append } 000\text{: } 110101\,000 \quad \leftarrow \textit{divisor has } n+1 \textit{ bits}$$
$$1101 \oplus 1011 = 0110 \rightarrow 110,\ \text{bring down } 0 \rightarrow 1100$$
$$1100 \oplus 1011 = 111,\ \text{bring down } 1 \rightarrow 1111$$
$$1111 \oplus 1011 = 100,\ \text{bring down } 0 \rightarrow 1000$$
$$1000 \oplus 1011 = 011,\ \text{bring down } 0 \rightarrow 0110 \quad \leftarrow \textit{leading 0: XOR 0000}$$
$$0110 \oplus 0000 = 110,\ \text{bring down } 0 \rightarrow 1100$$
$$1100 \oplus 1011 = 111 = \text{CRC} \quad \leftarrow \textit{quotient } 111101$$

(a) Codeword = **110101 111**. (b) $110101111 \div 1011$ gives remainder **000**, so it is accepted. (c) Received $110001111 \div 1011$ gives remainder **111** $\neq 0$, so the frame is rejected and (with ARQ) retransmitted. (d) A 3-bit CRC whose generator has more than one term detects **all single-bit errors**; since $x^3 + x + 1$ has a constant term it detects **all bursts of length** ≤ 3. It misses errors that are themselves a multiple of the generator.

PRACTICE Q10 HARD **[10 marks]**

Explain the hidden station problem in a wireless LAN and how CSMA/CA with RTS/CTS overcomes it. Use a timing diagram, and say why CSMA/CD cannot be used on Wi-Fi.

Answer

Hidden station: A and C are both in range of the AP B but out of range of each other. A senses the channel idle while C is sending, so both transmit and their frames **collide at B**; neither knows (draw the range figure in Section 4.2). **RTS/CTS** (draw the timing figure): after DIFS and a random back-off, A sends a short **RTS** with the time it needs. B replies **CTS**. C cannot hear A but **does hear the CTS**, so it sets its **NAV** and stays quiet. A sends DATA after SIFS; B returns an ACK after SIFS; then everyone contends again. Collisions can now only hit the short RTS frames, which are cheap to lose. **Why not CSMA/CD**: a radio cannot listen while sending (its own signal swamps the receiver), signals fade so a far collision is not noticed, and hidden stations never hear each other. So Wi-Fi **avoids** collisions and uses an **ACK** for every frame

instead of detecting them.

PRACTICE Q11 HARD **[10 marks]**

A computer lab has 24 PCs on three hubs (8 PCs each) chained together. Users complain of slow transfers. (a) How many collision and broadcast domains are there? (b) The hubs are replaced by switches, each switch connected to its own port on a router. Count the domains now. (c) Explain why performance improves and suggest how VLANs could be used. Draw the new design.

Answer

(a) Hubs repeat everything, so all 24 PCs are in **1 collision domain** and **1 broadcast domain**: lots of collisions and back-off. (b) Each switch port is a collision domain: 24 PC links +3 switch-to-router links = **27 collision domains**. Each router interface is a broadcast domain: **3 broadcast domains**. (Draw like the router panel in Section 8, with three switches.) (c) Switches send frames only to the destination port, give each PC dedicated full-duplex bandwidth and remove collisions (CSMA/CD no longer needed). The router stops broadcasts spreading between groups. With VLANs, PCs can be grouped by class or course across all three switches (e.g. VLAN 10 students, VLAN 20 lecturers), with 802.1Q trunks between switches and inter-VLAN routing on the router, giving security and smaller broadcast domains without recabling.

PRACTICE Q12 HARD **[10 marks]**

A 2 Mbps link is 3000 km long (signal speed 2×10^8 m/s) and carries 1000-byte frames. (a) Find the link utilisation of stop-and-wait. (b) What window size keeps the link busy? (c) How many sequence-number bits are needed for Go-Back-N and for Selective Repeat? (d) Which would you choose on a noisy link and why?

Answer

Working

$$T_{fr} = \frac{8000 \text{ bits}}{2 \times 10^6} = 4 \text{ ms}, \quad T_p = \frac{3 \times 10^6}{2 \times 10^8} = 15 \text{ ms}$$

$$a = 15/4 = 3.75, \quad U = \frac{1}{1+2a} = \frac{1}{8.5} \approx 11.8\% \quad \leftarrow \textit{(a)}$$

$$W \geq 1 + 2a = 8.5 \Rightarrow 9 \text{ frames} \quad \leftarrow \textit{(b) round up}$$

$$\text{GBN: } 2^m - 1 \geq 9 \Rightarrow m = 4 \quad \leftarrow 2^3 - 1 = 7 \textit{ too few}$$

$$\text{SR: } 2^{m-1} \geq 9 \Rightarrow m = 5 \quad \leftarrow 2^4 = 16$$

(d) **Selective Repeat**: on a noisy link many frames are lost; Go-Back-N would resend the lost frame *and every frame after it* (up to 8 extra frames each time), wasting bandwidth. SR resends only the damaged frame, at the cost of receiver buffers and more complex logic.

ONE-PAGE CHEAT SHEET

Layer 2 jobs: framing, MAC addressing, flow control, error control, access control. Node-to-node.

- **LLC** 802.2 (upper): SAP / protocol ID, flow, error control (ACK), media-independent
- **MAC** (lower): frame, 48-bit addresses, CRC, access (CSMA/CD, CSMA/CA, token)

Ethernet frame: Preamble 7 + SFD 1 | Dest 6 | Src 6 | Type 2 | Data 46–1500 | FCS 4. Frame 64–1518 B.
MAC address: 48 bits = OUI 24 (maker) + NIC 24. b0 I/G (uni/multicast), b1 U/L. Broadcast FF-FF-FF-FF-FF-FF. **ARP**: IP $\rightarrow$ MAC, broadcast request, unicast reply, cache.
ALOHA: pure $S = Ge^{-2G}$, vuln. $2T_{fr}$, max 18.4% at $G = 0.5$. Slotted $S = Ge^{-G}$, vuln. T_{fr}, max 36.8% at $G = 1$.
CSMA/CD (802.3, wired): sense, send, listen, collision $\rightarrow$ jam $\rightarrow$ back-off $R \in [0, 2^k - 1]$, $k = \min(n, 10)$, abort after 16. Min frame = $2T_p \times$ rate = 64 B.
CSMA/CA (802.11, wireless): DIFS + back-off, RTS/CTS, NAV, SIFS, ACK every frame. Fixes hidden station.
CD = cable, detect, after. **CA** = air, avoid, before.
Controlled access: reservation, polling, token. **FDDI**: 100 Mbps, dual counter-rotating fibre rings, token, fault tolerant.
802: .1 bridge/STP/VLAN(.1Q) .2 LLC .3 Ethernet .5 Token Ring .11 Wi-Fi .15 Bluetooth .16 WiMAX

Errors: single-bit (1 bit) vs burst ($\geq$ 2 bits, length first–last).

- **Parity**: even/odd count of 1s; misses 2 errors. 2-D parity locates 1 error.
- **CRC**: append n zeros, mod-2 divide by $(n + 1)$-bit generator, remainder = CRC; receiver remainder 0 = OK. Detects bursts $\leq n$.
- **Checksum**: 1's complement sum, wrap carry, send complement; receiver sum = all 1s.

Hamming distance = 1s in $x \oplus y$. Detect s: $d_{min} \geq s + 1$. Correct t: $d_{min} \geq 2t + 1$. Hamming code: $2^r \geq m + r + 1$, parity at 1, 2, 4, 8.
FEC = receiver corrects itself. **ARQ** = detect + resend.
Stop-and-wait: 1 frame, timer, ACK, resend on timeout, numbers 0/1. $U = 1/(1 + 2a)$, $a = T_p/T_{fr}$.
GBN: $W \leq 2^m - 1$, resend lost + all after. **SR**: $W \leq 2^{m-1}$, resend only lost, buffers.
Piggybacking: ACK rides on reverse data frame.
Devices: repeater/hub L1 (1 collision domain); bridge/switch L2 (1 collision domain per port, 1 broadcast domain); router L3 (1 broadcast domain per interface).
Bridge: learn source, filter, forward, flood, age; STP stops loops.
VLAN: logical broadcast domain on switches, 802.1Q tag (12-bit VID), trunk, inter-VLAN via router.